Yijie Ma

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Skills

C/C++, Go, Python, OpenTelemetry (OTLP, Semantic Conventions, W3C Trace Context), Distributed Systems, gRPC, AI & Agentic Observability (GenAI, MCP, A2A), Kubernetes, API Gateway/L7 Proxy, Networking, Storage & Filesystems, Desktop App Development (Linux/macOS/Windows)

Work Experience

Google April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Designed API Gateway's distributed tracing architecture across HTTP and gRPC requests, defined and implemented tracing data model based on OTel SemConv for Cloud, GenAI, Model Context Protocol (MCP), and Agent-to-Agent (A2A) requests.
- Built end-to-end W3C Trace Context propagation, trace span generation, trace context initiation and sampling, and asynchronous OTLP trace/log export pipelines for API Gateway handling millions of QPS.
- Led delivery of an isolated fast API serving release track that decouples rapid AI and Cloud feature rollouts from the slower stable release track.
- Pioneered reusable AI Skills to automate the release process, adopted by other team members.
- Mentored junior engineers via 1:1 architecture walkthroughs, milestone scoping, and code reviews to deliver multi-quarter observability projects.

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer of gRPC (40K+ GitHub stars) with 150+ authored PRs, driving open-source RFCs and PR reviews across C++ core, networking, and observability.
- Co-authored gRFC A79 to standardize non-per-call telemetry; architected the C++ templatized Metrics framework and OpenTelemetry plugin, cutting Counter overhead by 10%.
- Designed and built the cross-platform EventEngine asynchronous DNS resolver across Linux/POSIX and Windows.
- Designed the POSIX fork syscall handling system and resolved pre-fork deadlocks across gRPC C++ and Python.

Google Kubernetes Engine (GKE) Networking

- Engineered GKE control-plane automation in Go to auto-provision multi-NIC RDMA network topologies for AI-optimized GPU clusters.
- Designed concurrent request handling with distributed transactions and validated multi-node GPU communication using NVIDIA's NCCL test suite.

Google Drive Desktop Application

- Designed and built the rsync-style differential upload feature to save upload bandwidth for millions of users.
- Architected the USB device backup subsystem backing up millions of files daily, increasing backup speed 10x by optimizing I/O read buffers.
- Designed the cross-platform Preferences system for user and admin-level app configurations.
- Discovered a new category of crashes in threads running on heap-allocated stacks on Windows.

Stadia Cloud Gaming Service

- Designed a Linux Cgroup library for setting up container resource configurations on cloud gaming VMs.
- Increased game click-to-play launch speed by 75% (1.58s to 396ms) via CPU profiling and container setup optimization.

NetApp August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Designed and built a filesystem software to backup user's files to multi-cloud vendors. Designed and implemented full and incremental snapshot backup and restore functions using RocksDB.

Education

Carnegie Mellon University Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)